NISHANT GIRI

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Summary

Computer Science undergraduate and researcher with hands-on work in deep learning, computer vision, generative models, NLP, and reproducible machine-learning experimentation. Author of an accepted IEEE ICNPCV 2026 paper on GAN-based synthetic augmentation for data-scarce plant-disease classification, with experience building end-to-end research and software systems.

Education

Shri Ramdeobaba College of Engineering and Management, Nagpur — Expected 05/2027
B.Tech. in Computer Science and Engineering — CGPA: 8.57/10

Technical Skills

Programming: Python, Java, C, and JavaScript
Machine Learning: PyTorch, scikit-learn, XGBoost, Hugging Face Transformers, GANs, CNNs, and ensemble learning
Research: Experimental design, baseline comparison, data augmentation, ablation analysis, F1-score, precision, recall, and FID
Data and Development: NumPy, Pandas, FastAPI, REST APIs, MongoDB, PostgreSQL, SQL, React, Git, and GitHub

Research and Selected Projects

GAN-Based Synthetic Augmentation for Septoria Detection — 2025–2026
PyTorch, DCGAN, DiffAugment, FastGAN, Computer Vision — GitHub

- Designed a reproducible pipeline comparing DCGAN with DiffAugment and FastGAN for synthetic augmentation of a data-scarce Septoria leaf-image class.
- Evaluated real-only and synthetic-data experiments using FID and downstream classifier F1-score; analyzed whether augmentation improved generalization under limited training data.
- Authored the accepted IEEE ICNPCV 2026 paper *Enhancing Deep Learning Classifier Generalisation in Data-Scarce Domains through GAN-Based Synthetic Augmentation: A Study on Septoria Detection.*

InsightFlow: Multimodal Financial Intelligence — May 2026
Python, FastAPI, FinBERT, XGBoost, RAG, MongoDB — GitHub

- Built a machine-learning system combining Random Forest, XGBoost, and Gradient Boosting with engineered market indicators and walk-forward evaluation.
- Integrated FinBERT sentiment analysis, retrieval-augmented generation, speech-emotion inference, and persistent prediction-feedback storage.

Netra-Vani: Adaptive Assistive Communication — Feb 2026
Python, Computer Vision, Speech Recognition, K-means — GitHub

- Developed an assistive Morse-to-speech interface supporting mouse, voice, text-to-speech, and webcam-based eye-blink input.
- Applied K-means clustering and dynamic timing thresholds to adapt Morse-code decoding to individual tapping speed in real time.

Experience

AyLark — Dec 2025–Mar 2026
Frontend Developer — *Nagpur, India*

- Built responsive React and Next.js features, integrated REST APIs and PostgreSQL-backed workflows, and tested production features in Agile development cycles.

Additional Research and Coursework

Research: Co-developed and evaluated machine-learning models for soybean price forecasting in Maharashtra; manuscript submitted for publication.

Coursework: Data Structures and Algorithms, Artificial Intelligence and Machine Learning, Operating Systems, Computer Networks, DBMS, OOP, and Theory of Computation.